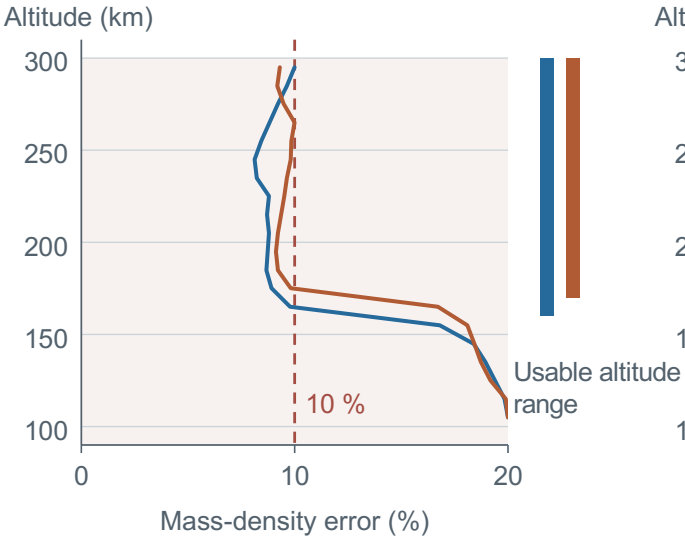
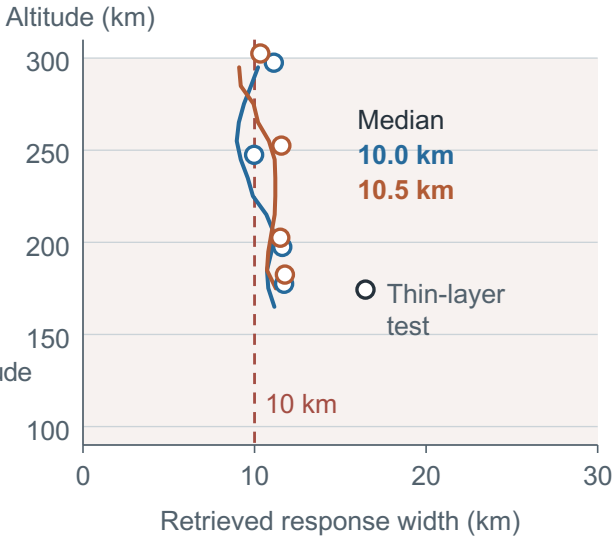
Modeled mass-density and temperature performance

One unit, one occultation — Quiet (ap 4) — Severe storm (ap 400) — Benchmark

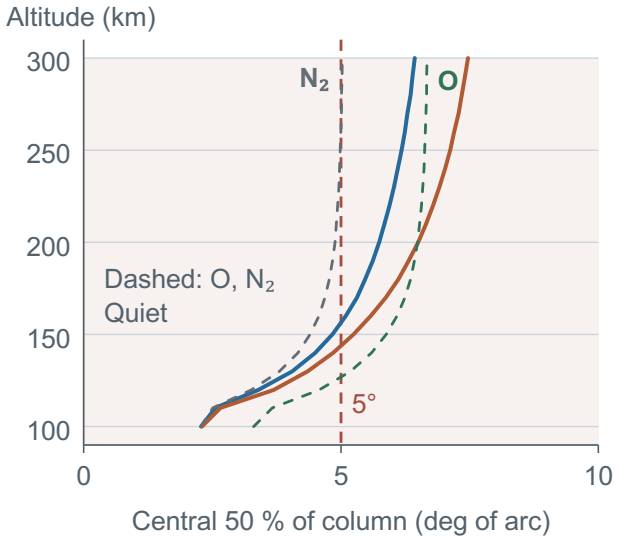
a Random error



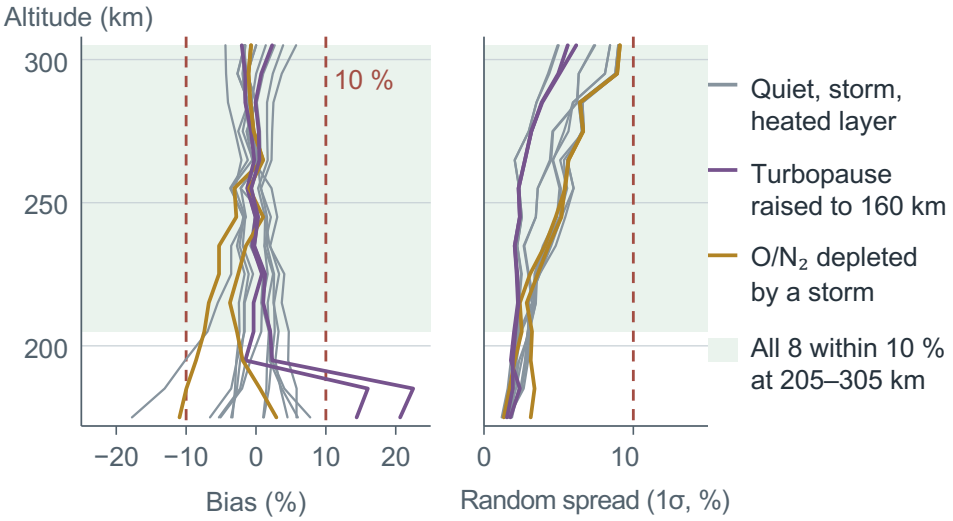
b Vertical resolution



c Horizontal resolution



d Temperature, eight test atmospheres



e Remote-sensing benchmark

Metric	Benchmark	Quiet	Severe storm
Altitude range	90–300 km	160–300 km	170–300 km
Random error	10 %	≤10 %	≤10 %
O / N_2 at 200 km	10 %	13 % / 16 %	13 % / 19 %
Vertical	10 km	10.0 km	10.5 km
Horizontal	5°	5.1–6.4°	5.9–7.5°
Retrieval bias		≤7.5 %†	≤9.8 %†
Calibration		≤2.5 % budget	≤2.5 % budget
Accuracy, total	**10 %**	**7.9 %**	**10.1 %**
Temperature	10 %, 10 km	≤10 %, 10.8 km‡	≤10 %, 10.8 km‡

† Bias: 7 of 8 atmospheres, 170–300 km; prior wrong by its stated error.
Raised turbopause: −11 to −16% bias below 182 km.
Total = root-sum-square of bias and ≤2.5% calibration.
‡ Temperature: bias and 1σ spread, 205–305 km.
O_2 has no row: above 200 km it follows the prior.

Figure 2. Modeled mass density uses measured Zr and In curves, blur and noise. Usable: ≤10% random error, and a true change recovered to within 20%. (d) Temperature: eight atmospheres, 30 noise draws each; priors vary O and N_2 by 45% and O_2 by 60%. Shading marks benchmark altitudes.